# HUY NGUYEN

SYSTEMS SOLUTION ENGINEER

work@huynguyen.email
huynguyen.space
linkedin.com/in/huy-nguyen-au
github.com/HuyNguyenAu

## OVERVIEW

Full stack software engineer building cloud-native services, edge systems, and web apps in C#/.NET, React/TypeScript, and Azure, owning them from design through to production. 6+ years of experience, including platforms powering national fleet electrification rollouts for Woolworths, RACV, SG Fleet, and Team Global Express.

## EXPERIENCE

**Full Stack Software Engineer**

JETCharge, Melbourne | Oct 2021 – Jun 2026

- **Multi-Tenant Billing (Illuminate):** Built an independent billing engine from scratch (ASP.NET Core, SQL Server, Stripe Connect split payouts) with per-session metering and time-of-use pricing, unlocking $100k - $300k/year in recurring revenue (measured) and eliminating manual billing disputes.
- **Customer Portals & Authorisation:** Built self-service asset delegation portals (React, TypeScript, Zustand, ASP.NET Core) and centralised multi-tenant authorisation in the Entity Framework data layer, removing duplicate permission checks and ensuring strict customer data isolation.
- **Remote Commissioning (ZTC):** Built a validated pre-configuration web tool and automated boot sync service, cutting on-site setup from 6h to 2 - 3.5h (zero with warehouse pre-pairing); saving an est. $325 - $520 per site in specialist field labour.
- **Cloud Performance & Decoupling:** Decoupled configuration schemas so web and firmware teams deploy independently; added a pointer-table cache in front of Blob Storage, slashing median device handshake from >30s to <3s and cutting service Azure costs by 5% - 10% (measured).
- **Edge Energy Management (CORE v2):** Built an offline-first C#/.NET daemon on Linux gateways balancing charger load in real time via Modbus and OCPP, enabling up to 5x more chargers per supply; built an embedded Next.js/WebSockets dashboard so electricians commission sites without engineers.
- **Delivery & Operations:** Maintained automated Azure DevOps CI/CD to per-environment Azure subscriptions with Bicep; wrote comprehensive test suites (xUnit, NUnit, MSTest, Jest, Vitest) with SonarQube and Roslyn analyzers in every pipeline; supported production on-call with Application Insights, Azure Monitor, and KQL.

## EDUCATION

**Computer and Network Engineering**

RMIT

Mar 2016 – Mar 2020

## SKILLS

**Backend:** C#, .NET / .NET Core, ASP.NET Core Web API, Entity Framework, REST APIs, Python

**Cloud:** Azure Functions, App Service, Container Apps, Service Bus, Event Grid, Bicep

**Data:** SQL Server / Azure SQL, PostgreSQL, Cosmos DB, Table and Blob Storage

**Frontend:** React, Next.js, TypeScript, Zustand, React Context, Vite, ShadCN UI

**DevOps & Quality:** Azure DevOps Pipelines, xUnit, NUnit, MSTest, Jest, Vitest, SonarQube / SonarCloud,

**Observability:** Application Insights, Azure Monitor, Log Analytics (KQL)

**Integrations:** Stripe Connect, WebSockets, KYC APIs, Modbus RTU / TCP, OCPP 1.6J / 2.0.1

- **Team & Stakeholders**: Ran collaborative design sessions with 3 - 5 engineers, led 1:1 code reviews and pair programming, onboarded juniors; worked in Scrum with POs and QA; gave technical architecture input to enterprise tenders.

**IT Specialist**
EnergyLocals, Melbourne | Oct 2020 – Oct 2021

- **Customer Onboarding:** Built the company's first white-label onboarding platform (Laravel, Bootstrap) with automated multi-tenant database provisioning and Westpac/Equifax KYC integration, cutting verification from 2 - 3 days to under 60s across 10,000+ customers.
- **Settlement Reconciliation:** Built automated daily reconciliation in Python and SQL between AEMO wholesale electricity settlements and customer meter data, plus billing calculation pipelines for time-of-use and wholesale pass-through tariffs.
- **Hardened Internal Operations:** Administered high availability Linux VPS database servers, authoring automated Bash and PowerShell disaster recovery pipelines and workstation provisioning scripts that cut onboarding from 2 days to under 2 hours.

**Graduate IT Specialist**
EnergyLocals, Melbourne | Sep 2019 – Oct 2020

- **Market Data Automation:** Automated daily verification of AEMO wholesale settlement runs (Python, SQL), identifying calculation discrepancies early with zero unresolved variances, and built ingestion feeds for electricity spot prices and regional demand data.
- **Systems Automation:** Built automated database backup routines and disaster recovery procedures (PowerShell, Bash, Python) to safeguard customer records, and standardized developer workstation setup scripts.

## SYSTEMS & APPLIED AI RESEARCH (OPEN SOURCE)

- **Wayfare (C#):** A coding agent bounded by a finite state machine to make agentic workflows reproducible, predictable, and easy to debug.
- **LLM Native Virtual Machine (Rust):** An assembly-like virtual machine that breaks prompts into small, predictable atomic steps to improve accuracy for small local models.
- **Lox Lang AI (C#):** A bytecode interpreter and AST evaluator used to explore AI-generated and AI-optimised execution routines.
- **GPT Decoder (Python):** A minimal language model engine built from scratch to test how models run on local hardware without depending on third party cloud providers.
- **Wobbly ML (Python):** A lightweight machine learning library built from first principles to explore backpropagation and numerical gradients.